

How to Publish in Q1 Journals and Top-Tier Conferences?

Comprehensive step-by-step guideline for preparing and publishing research

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Day 2

CRITICAL ANALYSIS

Critical Analysis

- Evaluates the credibility and quality of research evidence.
- Examines the methodology, relevance, limitations, and contribution.
- Checks whether conclusions are supported by the data.
- Compares findings with existing studies.
- Goes beyond simply summarizing the paper.

Critical Analysis

☐	Is the study methodologically sound?	Ensures validity of the results.
☐	Are the findings well-supported by data?	Avoids overgeneralization or unsupported claims.
☐	Does the paper align with other studies?	Helps understand consistency or divergence.
☐	What are the limitations or biases?	Identifies what's missing or weak.
☐	How is this study relevant to my research?	Keeps your review focused and purposeful.

Critical Analysis



Critical Analysis



Writing a Coherent Literature Review

A strong literature review should present studies in a clear, connected, and analytical sequence, rather than discussing papers individually.

1. Organize the Literature by Themes

Group studies according to common topics, methods, findings, or debates. Studies on AI in education can be grouped into three main themes:

Writing a Coherent Literature Review

2. Move from General to Specific

Begin with the broader research field, then gradually narrow the discussion to the specific research problem.

Example

Machine learning has been applied in healthcare, finance, and education. Within education, recent studies have examined how AI can support personalized learning and improve student outcomes.

Writing a Coherent Literature Review

3. Use Linking and Contrast Phrases

Connect studies and show relationships between their findings.

Common linking phrases:

- Similarly...
- In contrast...
- However...
- Furthermore...
- Consistent with...
- Unlike previous studies...
- Taken together...

Writing a Coherent Literature Review

4. End with Synthesis, Not Summary

Conclude each section by explaining the overall evidence, limitations, and implications.

Example

Taken together, the studies suggest that AI can improve learning outcomes, but limited dataset diversity and insufficient real-world validation remain important challenges.

EXPERIMENT

Experiment

1. Establish a Baseline

A baseline is a simple or established method used as a reference point for evaluating the proposed approach.

- Relevant to the research problem
- Commonly used in previous studies
- Reproducible using available data and code
- Evaluated using the same dataset, data split, and metrics

Experiment

2. Reproduce the Baseline

Where to Find the Code

1. Check the paper for GitHub, project-page, or supplementary links.
2. Search the paper title on GitHub or Papers with Code. 
3. Implement the method from the paper if no code is available.
4. Contact the authors for missing code or details.

Experiment

3. Reverse-Engineer the Baseline

☐	Reproduce the baseline	Confirm the original method and performance.
☐	Understand the pipeline	Study the model architecture and training process.
☐	Identify key choices	Determine the main components, assumptions, and design decisions.
☐	Perform ablation studies	Modify or remove one component at a time.
☐	Analyze the results	Explain why each component improves or reduces performance.
☐	Propose an improvement	Use the findings to design a justified modification.

Experiment

4. Document and Organize

- Record experiments, decisions, questions, and failures as they occur.
- Use written notes or short voice notes with dates and timestamps.
- Store code, results, papers, and notes in clearly named folders.
- Back up your work using cloud storage and version control.
- Maintain an experiment log containing settings, results, and observations.

Experiment

 **Research Experiment Logs**



How to Find a Research Solution

- Revisit recent papers to explore alternative methods and unresolved limitations.
- Discuss the problem with supervisors, researchers, and team members.
- Generate several possible improvements based on evidence from the literature.
- Test one change at a time using controlled and fair experiments.
- Compare the results with the baseline and refine the most promising solution.

Key Terms in a Research Contribution

Research Gap	What has not been adequately addressed?
Proposed Solution	How will the study address the problem?
Novelty	What is new or different from previous studies?
Contribution	What new knowledge, method, evidence, or resource will the study provide?
Significance	Why is this contribution important?
Limitations	What does the study not fully address?

Experiment Design

☐	Dataset	What data will be used, and is it suitable for the research problem?
☐	Data Preparation	How will the data be cleaned, transformed, and divided?
☐	Baseline Methods	Which existing methods will be used for comparison?
☐	Proposed Method	What new model, framework, or improvement will be tested?
☐	Evaluation Metrics	How will performance be measured?
☐	Experimental Settings	What training settings, hardware, and parameters will be used?
☐	Ablation Study	Which components contribute to the final performance?
☐	Robustness Testing	Does the method remain effective under different conditions?
☐	Error Analysis	Where and why does the method fail?
☐	Statistical Validation	Are the improvements consistent and statistically meaningful?

Experiment Design



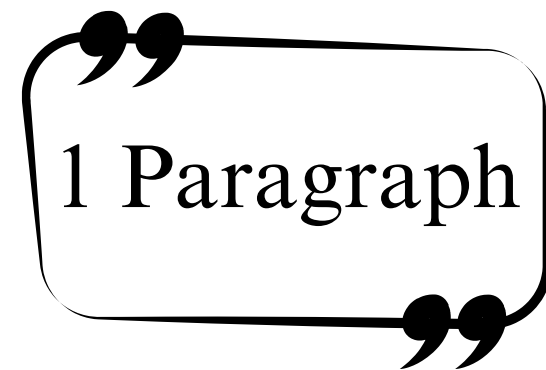
Experimental design Sheet



PAPER WRITING

Paper Writing

A strong academic paragraph should focus on one central idea and develop it **logically**.



=



1 idea

Paragraph Formula

Main idea → Evidence → Explanation → Conclusion

Paper Writing

Academic writing should communicate ideas **clearly** and **directly**, using only the words necessary to express the intended meaning.

In order to evaluate	To evaluate
At this point in time	Currently
The results were very good	The model achieved 92% accuracy

Paper Writing

Academic claims should be supported by reliable evidence, such as research findings, data, or established theory. Sources must be cited clearly.

AI models are always accurate.	AI models have achieved high accuracy on several benchmark datasets [1].
Many researchers believe this method is effective.	Previous studies reported improved performance using this method [2,3].
This problem is very common.	The problem affects approximately 30% of the studied population [4].
Copying an idea without citation	Paraphrasing the idea and citing the original source

Paper Writing

Academic writing often uses an objective and impersonal style. Avoid first-person expressions such as “I think,” “I believe,” or “in my opinion”.

Use expressions such as this study, this research, the proposed method, the results, or the findings instead of we and our.

Paper Writing

Established facts and general knowledge	Present tense	Deep learning is widely used for image classification.
Theories and accepted concepts	Present tense	The theory explains how users adopt new technologies.
Existing literature	Present or present perfect	Previous studies show that the method is effective. / Researchers have investigated this problem.
Research gap	Present or present perfect	Few studies have examined this issue in real-world settings.
Aim and objectives	Present tense	This study aims to evaluate the proposed method.

Paper Writing

Methodology	Present tense	The model takes images as an input.
Experiments performed	Past tense	The model was trained for 50 epochs.
Results of the current study	Past tense	The proposed model achieved 92% accuracy.
Interpretation of results	Present tense	These findings indicate that the method is robust.
Figures and tables	Present tense	Table 2 shows the comparison between the models.
Conclusion	Present or past tense	The study demonstrates that the method is effective. / The model outperformed the baselines.
Future work	Future tense	Future research will examine larger and more diverse datasets.

ABSTRACT

Abstract

☐	Background / Context	Briefly introduce the topic and explain why it is important.	Use one or two sentences to establish the context.
☐	Research Problem / Objective	State the specific problem, research gap, or main objective.	Be clear, focused, and direct.
☐	Methods / Approach	Briefly describe the data, method, model, or experimental procedure.	Include only essential methodological details.
☐	Results / Findings	Report the most important findings.	Use specific results rather than vague statements.
☐	Conclusion / Implications	State the main conclusion, contribution, and potential impact.	Explain why the findings matter.

Abstract

Abstract

Despite the central role of optimization in deep learning, most optimizers rely on update structures whose functional form is fixed before training begins. This static design can limit their ability to respond to changing gradient behavior across the loss landscape, where training may shift between stable, noisy, and inconsistent regimes. This study proposes PILOT (Policy-Informed Learned OpTimizer), an online optimizer that adapts its update behavior during training. Rather than using a fixed balance between momentum, normalization, and sign-based updates, PILOT uses gradient-direction agreement as a signal of local training stability. Conditioning the update rule on this agreement signal allows the optimizer to adjust its behavior when gradients become stable, noisy, or inconsistent. Experiments on FashionMNIST and CIFAR-10 show that PILOT consistently achieves the highest accuracy among the evaluated optimizers across convolutional settings. On the CNN architecture, PILOT reaches 94.13% on FashionMNIST and 81.94% on CIFAR-10. On ResNet-18, it further improves performance, reaching 95.71% on FashionMNIST and 93.42% on CIFAR-10. These results suggest that learning how to adapt the update structure during training can improve performance across both compact and deeper convolutional models while preserving a simple first-order optimization framework. The implementation of PILOT is publicly available at <https://github.com/SattamAltwaim/PILOT.git>.

Background

Problem

Problem

Results

Conclusion

INTRODUCTION

Introduction

☐	General Background / Context	Introduce the broader research field and explain why the topic is important.	Begin broadly, then gradually narrow the discussion.
☐	Problem Statement / Research Gap	Identify the specific problem, limitation, or unresolved issue in existing research.	Clearly explain what is missing and why it matters.
☐	Motivation	Explain why the problem deserves attention and what motivates the study.	Highlight the need, urgency, impact, or limitations of current approaches.
☐	Research Aim / Objectives	State what the study intends to investigate, develop, evaluate, or achieve.	Use precise, focused, and action-oriented language.
☐	Significance / Contribution	Explain how the study will add knowledge, improve a method, or provide practical value.	Emphasize the scientific or practical importance of the work.
☐	Paper Structure Optional	Briefly outline the organization of the remaining sections.	Keep it concise and include it only when required by the venue.

METHODOLOGY

Methodology

☐	Model Overview	Introduce the proposed model and explain its main objective.	Provide a high-level view of the architecture and how the main components work together.
☐	Model Architecture and Components	Describe the main modules, layers, and internal structure of the model.	Organize the discussion into subsections covering the encoder, feature extraction layers, hidden blocks, and prediction head.
☐	Input, Output, and Data Flow	Define the model inputs and outputs and explain how data moves through the architecture.	Include input format, dimensions, preprocessing, intermediate representations, and final predictions.
☐	Mathematical Formulation and Training	Present the equations that describe the model operations and optimization process.	Define all variables and explain the transformations, loss function, optimizer, and main training settings.

RESULTS

Results

- Present the main findings clearly and objectively.
- Use tables, figures, or charts when they improve understanding.
- Highlight only the most important patterns and comparisons.
- Report exact values when necessary.
- Avoid repeating every number already shown in a table or figure.

DISCUSSION

Discussion

- Explain what the results mean.
- Relate the findings to the research question and objectives.
- Compare the results with previous studies.
- Explain possible reasons for agreement or disagreement.
- Discuss unexpected or negative findings.
- Highlight the significance and contribution of the study.

CONCLUSION

Conclusion

- What research problem did the study address?
- What was done to investigate the problem?
- What were the main results?
- What is the main contribution of the study?
- Was the research question answered?
- What are the main limitations?
- What should future research investigate or improve?

PUBLICATION

Types of Academic Publications

Journal Articles

- Peer-reviewed scholarly papers.
- Typically longer and more detailed.
- Aim for indexed journals (Scopus, Web of Science).
- Examples: Q1, Q2 journals based on impact factor.

Types of Academic Publications

Conference Papers

- Shorter papers presented at academic conferences.
- Often faster publication process.
- May include oral presentations or poster sessions.
- Good for networking and getting feedback.

 [AI Conference Deadlines](#)

Types of Academic Publications

Book Chapter

- Research or review content published as part of an edited academic book.
- Providing detailed discussion of a specialized topic or established body of work.

Types of Academic Publications

Workshop Papers / Proceedings

- Shorter or early-stage studies presented at focused workshops, often alongside major conferences.
- Exploring emerging ideas, preliminary results, and specialized research topics

Types of Academic Publications

Preprints

- Publicly available manuscripts shared before formal peer review, commonly through platforms such as arXiv or SSRN.
- Rapidly sharing research findings and receiving early community feedback

Editorial Decision

Accept	The paper is accepted for publication, usually with only final formatting or proofreading changes.
Minor Revision	The paper requires limited corrections, such as improving clarity, language, references, or presentation.
Major Revision	The paper requires substantial changes to the methodology, experiments, analysis, interpretation, or structure before reconsideration.
Reject and Resubmit	The current submission is rejected, but the authors may prepare a substantially revised version and submit it as a new manuscript.
Reject	The paper will not be considered further because of issues such as weak contribution, methodological flaws, insufficient evidence, or mismatch with the venue's scope.

Reviewer Response

- Thank the reviewer for the comment or suggestion.
- Acknowledge the concern and show that it has been understood.
- Respond clearly by either revising the manuscript or providing a reasoned clarification.
- Describe the change and indicate where it appears in the revised paper.
- Support the response with evidence, additional experiments, or references when necessary.

RESEARCH JOURNEY

Reviewer Response

Challenges

- Maintaining motivation during long or unsuccessful experiments
- Managing time and competing responsibilities
- Handling criticism, rejection, and repeated revisions
- Remaining patient when progress is slow

Reviewer Response

Advantages

- Critical thinking and problem-solving
- Patience, discipline, and subject mastery
- Objectivity and evidence-based reasoning
- Teamwork and collaboration
- Continuous learning and adaptability
- Scientific communication and professional confidence

Reviewer Response

Tips

- Cultivate curiosity.
- Continuous learning.
- Read extensively.
- Regularly seek guidance from supervisors and mentors.
- Help and learn from your colleagues.

Q&A

THANK YOU